

FRE-GT-9743 Assignment 2

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xVA

Bond Forward with Daily VM and Funding Benefit (Computable)

Bank A and Client B enter into a bond forward at $t = 0$. Bank A is long the forward. The trade is collateralized with daily variation margin (VM) in cash under the following rule:

- Let V_t be the end-of-day (EOD) mark-to-market of the forward from Bank A's perspective (positive means Bank A is in-the-money).
- The collateral account is updated each day to satisfy

$$C_t = V_t$$

(full collateralization, zero threshold, zero MTA).

- The VM call paid at EOD of day t is

$$\Delta C_t = C_t - C_{t-1}.$$

- Cash collateral posted earns the collateral rate

$$r_C = 4\% \quad (\text{simple interest, ACT/360}).$$

Notional is \$10,000,000 and the forward settlement date is 6 months later (the settlement date is not needed for the calculations below).

At Day 1 EOD, the forward is at par:

$$V_1 = 0 \quad \Rightarrow \quad C_1 = 0.$$

1. On Day 2 EOD, the MTM becomes

$$V_2 = -200,000.$$

Which party posts VM at Day 2 EOD, and what is the VM amount ΔC_2 ?

Ans: $\because C_2 = V_2 = -200,000 \therefore \Delta C_2 = C_2 - C_{2-1} = -200,000 - 0 = -200,000 \therefore$
Bank A pays \$200,000 variation margin at Day 2 EOD.

2. Alternatively, suppose on Day 2 EOD the MTM is

$$V_2 = +150,000.$$

Assume Bank A can invest any received cash at its unsecured funding/investment rate

$$r_F = 6\%.$$

Compute the funding benefit over Day 2 (one day of interest).

Ans: On Day 2 Client B should post Bank A \$150,000, and invest at a rate of 6% for one day, therefore, A can get funding benefit $150,000 \times 6\% \times 1/360 = \25

Exposure Calculation for a Bond Forward

The key component in calculating xVA is the exposure calculation.

Suppose we long a bond forward contract that:

- has notional N (e.g., $N = 1$ million),
- settles at T (e.g., $T = 1Y$),
- strikes at K (e.g., $K = 100$).

At intermediate time $t \in (0, T)$, we shall calculate:

$$E(t) = \max\{V(t), 0\}.$$

Here, we assume zero collateralisation, and $V(t)$ is the valuation of the bond forward at time t .

Q1.

In Assignment 1, you worked out the carry method to price a bond forward contract:

- Repo the spot bond ($t = 0$) to settlement date T at repo rate r_R ;
- Subtract the coupon reinvested at repo rate r_R paid between 0 and T ;
- Discount the payoff at unsecured funding rate r_F .¹

For exposure calculation, we need to perform the above steps at time t . This requires simulating some relevant dynamics.

Claim: we need at least three stochastic factors X_1 , X_2 , and X_F that enable separating:

¹Notice, here we assumed no collateralization; otherwise, it will be discounted at rate r_C .

- bond yield,
- repo rate,
- unsecured funding rate.

Give your justification of this claim.

Ans: At time t , valuing the bond forward via the carry approach requires the bond price $B(t)$ depending on bond yield which is a stochastic term structure.

The financing from t to T , which depends on the repo rate $r_R(t, \cdot)$ and coupon reinvestment at repo.

And discounting of the forward payoff, which uses the unsecured funding which uses the unsecured funding curve r_F .

Besides, X_1 , X_2 and X_F are not identical, and can move with different spreads and imperfect correlations, a model with fewer than three stochastic drivers would force unrealistic deterministic relationships. Therefore we need at least three factors to separately model yield, repo, and funding dynamics.

Q2.

Let us adopt a multi-factor Gaussian model, in which case an i -discount factor has closed form:

$$P^i(t, T) = \exp \left(A^i(t, T) - B^i(t, T)^\top X(t) \right),$$

and $X(t)$ follows some known dynamics (Ornstein–Uhlenbeck process), thus A^i and B^i are all known.

- Is $P^i(t, T)$ known at time t or at time T ?
- Can you express $V(t)$ in terms of $P^i(t, \cdot)$?

Ans: Since $X(t)$ is known at time t and A and B are all known, thus $P^i(t, T)$ is known at time t not at time T .

If bond pays coupon at time T_k , then

$$B(t) = \sum_k c_k P^y(t, T_k) + P^y(t, T_n)$$

At maturity T : $payoff = B(T) - K$.

Then at time t , $V(t) = P^F(t, T) \mathbb{E}_t[B(T) - K]$

Under risk-neutral method, this can simplified to

$$V(t) = B(t) - K P^F(t, T)$$

and plug $B(t)$ and $P^I(t, T)$ into this we get

$$V(t) = \sum_{j=1}^m c_j \exp \left(A^y(t, T_j) - B^y(t, T_j)^\top X(t) \right)$$

$$+ \exp\left(A^y(t, T_m) - B^y(t, T_m)^\top X(t)\right) - K \exp\left(A^F(t, T) - B^F(t, T)^\top X(t)\right)$$

Q3.

Following the notes, finish the steps to compute CVA for a long position in this contract. (This is to make sure you understand the procedures to calculate xVA based on a given exposure profile.)

Ans: Choose a time grid $\{t_i\}_{0 \leq i \leq n}$. And choose K paths, I prefer k=5,000. Use Monte Carlo Simulation, to simulate state factors $\{X_t\}$. Here should include three stochastic factors mentioned in Q1.

Compute discount factors via closed form:

$$P^i(t, T) = \exp\left(A^i(t, T) - B^i(t, T)^\top X(t)\right),$$

Compute the forward value $V(t)$ following final answer in Q2.

Compute pathwise exposure:

$$E^k(t_i) = \max(V(t_i), 0) = V(t_i)^+$$

Then estimate expected exposure at each time point:

$$EE(t_i) = \frac{1}{M} \sum_{k=1}^M E^{(k)}(t_i)$$

Next step is specify default model. Let counterparty default time be τ and hazard rate $\lambda(t)$,

$$S(t) = \mathbb{Q}(\tau > t) = \exp\left(-\int_0^t \lambda(u) du\right),$$

$$dPD(t) = \mathbb{Q}(t < \tau \leq t + dt) = \lambda(t)S(t)dt$$

On a grid, $\Delta PD_i \approx \lambda(t_i)S(t_i)\Delta t_i$

Choose a recovery rate R and let discount factor $D(0, t)$ be discount factor.

Finally, the CVA can be gotten by

$$CVA = (1 - R) \int_0^T D(0, t) EE(t) dPD(t)$$

$$= (1 - R) \int_0^T D(0, t) EE(t) \lambda(t) S(t) dt$$

. In simulation — discrete grid approximation,

$$CVA \approx (1 - R) \sum_{i=0}^n D(0, t_i) EE(t_i) \Delta PD_i$$

Q4.

As a follow-up question: how would you come up with a sensible hazard rate that is factored into your survival probability calculation?

(Suppose you're buying from a bank with a certain credit rating.)

Ans: 1) Since $S(t)$ can be calculated as $S(t) = \exp(-\int_0^t \lambda(u) du)$. So on a grid

$$\lambda_i \approx -\frac{\ln S(t_i) - \ln S(t_{i-1})}{t_i - t_{i-1}}$$

2) If the counterparty has liquid CDS or liquid bonds, so we can infer a hazard rate from credit spreads

$$s \approx (1 - R)\lambda \Rightarrow \lambda \approx \frac{s}{1 - R}$$

3) Another approach is starting with rating-based historical PD curves to find $\lambda_{hist}(t)$. Adjust to current market conditions using a sector spread level.

Sub-Class and Delegation

Part I: Date Utilities

The `date` folder provides a set of utility functions related to date handling. You are expected to familiarize yourself with these date-related functions and may test or debug them using the provided test code.

No code implementation is required in this part.

Part II: Linear Products

The code templates for:

- Bullet Cashflow
- Overnight Index

will be provided.

Students are required to complete the implementations for:

- Interest Rate Stream
- Swap

After completing the required code, you should be able to display the full information of the corresponding financial products.

If you are not familiar with swaps, please review the relevant past lectures on Notion before starting this part.

Part III: Option Strategy Registry

Please use the singleton class to finish the registry of the `OptionStrategyRegistry`.